Job-Skill Matching with Skill Type Classification*

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Abstract

This literature review addresses the task of Job-Skill Matching with Skill Type Classification. The task requires, given a job title, retrieving a ranked list of relevant skills and classifying each as core, complementary, or transversal. We are provided a training set of at least 5,000 job title–skill pairs, a development set of 200 job titles with ESCO-normalised skill annotations carrying a core/contextual relevance label, and a test set of 500 job titles for blind evaluation. We survey six prior works that address skill extraction, weakly supervised skill matching, occupational classification, domain-adaptive pretraining, technical NER, and transversal skill detection — each contributing methods or findings we draw on in building our approach.

Keywords

job-skill matching, skill type classification, ESCO, transformer models, contrastive learning, named entity recognition, information retrieval

1. Introduction to the Task

The task of Job-Skill Matching with Skill Type Classification sits at the intersection of information retrieval and multi-class labelling. A system must take a raw job title as input, retrieve a ranked list of skills from the ESCO taxonomy, and simultaneously assign each retrieved skill one of three labels: *core*, *complementary*, or *transversal*. The training data offers at least 5,000 job title. The development set introduces a richer annotation layer of 200 job titles with ESCO-normalised skills marked as either core or contextual, which forces a model to capture the difference between what a job *always* demands and what it *sometimes* demands. The challenge lies in combining semantic retrieval with skill-type classification under ranking-based evaluation metrics such as nDCG and MAP. Because TalentCLEF includes multilingual data, models must operate across languages without relying solely on translation. Additionally, skill labels are structured and taxonomy-aligned, requiring systems to distinguish between different types of competencies rather than simply detecting skill mentions.

2. Literature Review

The work most relevant to this task covers three areas: pulling skills out of job text, sorting those skills into the right categories, and dealing with skills that are vague or implied rather than stated outright. Six papers are reviewed below, each approaching one or more of these problems from a different angle.

Decorte et al. [1] build a two-stage classifier that detects transversal skills in job ads using the ESCO taxonomy as the label standard. The first stage assigns a skill to one of six broad categories—Core, Thinking, Self-management, Social/communication, Physical, and Life skills. The second stage breaks that category down further into subclasses. Both stages use 768-dimensional sentence embeddings produced by a BERT-based Sentence Transformer fed into neural classifiers. One result worth highlighting: a multilingual embedding model performed within 0.55% accuracy of an English-only model. That narrow gap is encouraging for our task, since it means we may not need a separate translation

CLEF 2025

*Working notes paper submitted to CLEF 2025.

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step to handle non-English job titles. The paper's label structure also maps directly onto the core/complementary/transversal distinction in TalentCLEF, making it a natural reference for how we set up the classification layer.

Gugnani and Misra [2] take a different approach to the extraction problem. Instead of manually labelling training data, they use ESCO itself to generate labels automatically. The idea is simple: pull all n-grams of length one to four from a job posting, embed them, and compare each embedding against the embeddings of ESCO skill entries using cosine similarity. Any n-gram that clears a set threshold gets labelled as a skill. They test three ways of building these embeddings on top of both RoBERTa and the job-domain model JobBERT—encoding the phrase on its own (ISO), averaging its embedding across every time it appears in the corpus (AOC), and weighting token embeddings by IDF scores (WSE). WSE comes out on top, hitting a loose F1 of 59.61 with RoBERTa on the Sayfullina benchmark. On the harder SkillSpan benchmark, however, JobBERT pulls ahead of RoBERTa, which confirms that pre-training on job-specific text makes a real difference.

JobBERT-V3, presented by Decorte et al. [3], is trained on 21 million job titles across English, German, Spanish, and Chinese using GPT-4.1-generated synthetic translations of a proprietary English dataset. The model is built on multilingual MPNet and projects its 768-dimensional representations up to 1024 dimensions through an asymmetric linear layer. It is trained with InfoNCE contrastive loss, where shared ESCO skill annotations act as the bridge that pulls equivalent job titles from different languages toward the same point in embedding space. When applied to skill retrieval without any task-specific fine-tuning, it reaches a MAP of 0.245 on the validation split.

Bhola et al. [4] train SkillBERT, a BERT model trained from scratch on 700,000 job requisitions covering 2,997 unique skills, specifically to produce useful skill embeddings. The downstream task is to classify each skill into one or more of 40 competency groups, making it a multi-label problem. Their feature set combines three types of signal: cosine similarity scores between a skill's embedding and each group centroid, cluster membership labels generated by K-means (45 clusters) and spectral clustering (35 clusters), and the output probabilities from a two-layer neural network classifier trained on the SkillBERT embeddings directly. The results show a clear jump in XGBoost F1 when moving from pre-trained BERT (60.84%) to Word2Vec (66.73%) to SkillBERT (90.35%). The main takeaway is that the quality of the embedding model matters far more than the cleverness of the features built on top of it. For us, this is a strong case for starting from a job-domain checkpoint rather than a general BERT, even if training a full SkillBERT from scratch is not feasible here.

Pooja et al. [5] focus on extracting technical skills from data science job postings scraped from Indeed, covering 6,325 vacancies. They compare TF-IDF, text classification, and Named Entity Recognition (NER) as candidate methods, settling on NER because it reads context and grammatical structure rather than just counting word frequencies. They fine-tune three pre-trained models: spaCy, BERT in its RoBERTa form, and XLNet. BERT gives the best overall result with an F1 of 93.17%, supported by a precision of 92.31% and a recall of 94.19%. XLNet hits the highest precision at 95.22% but recall falls to 69.96%, which is a meaningful trade-off when missing a skill costs more than a false alarm. The paper also makes a practical point about preprocessing: dropping stopwords and applying stemming, steps that help traditional models, actually hurt transformer-based models because transformers need the full sentence to build accurate contextual representations. That finding directly shapes the preprocessing choices we make.

Vera et al. [6] build an occupational classifier for the Chilean labor market, mapping job postings to codes in the four-level CIUO-08/ISCO-08 taxonomy. Their dataset of 4,605 postings, drawn from six online job boards and labeled by domain experts The model is BETO, a Spanish-language BERT, fine-tuned on this data. At the most detailed (four-digit) level it reaches a weighted accuracy of 0.69 and an F1 of 0.66, beating both an SVM and an LSTM baseline. The SVM holds up reasonably at broad levels but falls apart at the fine-grained ones; the LSTM underperforms across the board. The main structural issue this paper surfaces is cascading errors: if a prediction is wrong at the top level of a hierarchy, every level below it is also likely to be wrong. Our three-way classification (core/complementary/transversal) does not have four levels, but core is implicitly the most selective of the three labels, and the lesson here is that models which account for that structure tend to do better than those that treat all labels as

equivalent and independent.

Across all six papers, a few consistent takeaways emerge. Domain-adapted encoders reliably beat general-purpose BERT for both extraction and classification. Cosine similarity against ESCO embeddings is a more robust retrieval signal than keyword matching. Transversal skills need their own dedicated treatment and do not respond well to the same NER pipelines used for technical skills. And benchmark scores should be read with some caution, since label noise in evaluation sets is a documented problem in this space.

References

[1] J.-J. Decorte, J. Van Hautte, T. Demeester, C. Develder, Techniques for transversal skill classification and relevant keyword extraction from job advertisements, Information 16 (2025) 167. URL: https://www.mdpi.com/2078-2489/16/3/167. doi:10.3390/info16030167.

[2] A. Gugnani, H. Misra, Implicit skills extraction using document embedding and its applications in job description parsing, in: arXiv preprint, 2020. URL: https://arxiv.org/pdf/2209.08071. doi:10.48550/arXiv.2209.08071.

[3] J.-J. Decorte, et al., JobBERT-V3: A contrastive learning model for cross-lingual job title matching, arXiv preprint arXiv:2507.21609 (2025). URL: https://arxiv.org/pdf/2507.21609v1.

[4] A. Bhola, K. Halder, A. Prasad, M.-Y. Kan, SkillBERT: "skilling" the BERT to classify skills, in: Proceedings of the 28th International Conference on Computational Linguistics, 2020. URL: https://www.studocu.com/in/document/sardar-vallabhbhai-national-institute-of-technology-surat/integrated-masters-in-physics/1-skillbert-skilling-the-bert-to/86064638.

[5] e. a. Pooja, Technical skills information extraction from resumes using advanced natural language processing model with transfer learning, Research Square Preprint (2024). URL: https://www.researchsquare.com/article/rs-7744440/v1. doi:10.21203/rs.3.rs-7744440/v1.

[6] e. a. Vera, A transformer-based intelligent system for hierarchical occupational classification in the labor market, IEEE Access (2025). URL: https://ieeexplore.ieee.org/document/11250910. doi:10.1109/ACCESS.2025.11250910.

A. Online Resources

The sources for the ceur-art style are available via

- GitHub,
- Overleaf template.